How AdamW and Muon Train a Small Language Model

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Summary. We train a small language model with AdamW and Muon from one initialization, on the same mini-batches. (1) Muon trains faster, and on a slightly different path. The final models are about as far apart as two data orders, but Muon's loss is lower. (2) The optimizers differ in various metrics that look at the model's internals: weight spectra, similarity of consecutive updates, curvature, and MLP sparsity. (3) When a run switches optimizer mid-training, its updates change at once. For its weight spectra and MLP sparsity, the earlier the switch, the closer they end to the second optimizer's own run; the same holds for sharpness after a switch to Muon. Even a switch at step 200 leaves a trace.

Setup. *Model.* We train a 14M-parameter nanoGPT (Karpathy, 2022) on TinyStories (Eldan and Li, 2023) for 10,000 steps of 128 sequences of 256 tokens (Appendix A lists the model, data, training and optimizer settings).

Optimizers. Muon (Jordan et al., 2024) orthogonalizes the update of each weight matrix in the transformer blocks: it sets the update's singular values to about 1. AdamW trains Muon's other parameters. Each optimizer uses the best of five learning rates. All runs start from the same weights. Each optimizer runs on the same five *data orders*.

Step 1: how the trajectories differ (Fig. 1). *Comparing runs.* The distance between two runs is the mean Jensen–Shannon divergence (JS, in bits) of their next-token predictions on held-out text. We compare runs at equal step and at equal training loss; at equal loss, speed does not count as a difference. The *floor* is the mean distance between two data orders of one optimizer. A distance of one floor means "as different as two data orders".

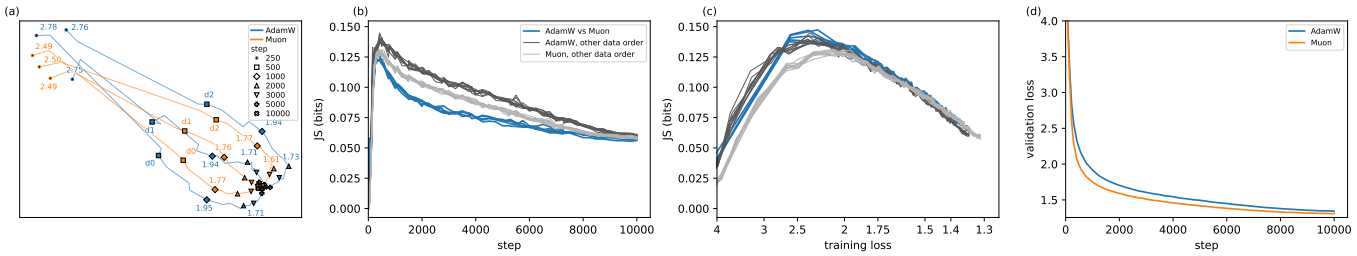


Figure 1: AdamW and Muon. (a) Checkpoints from step 250 on, mapped to 2-D by densMAP (shows neighbours, not distances); labels: training loss and data order. (b) JS between runs at equal step. (c) The same at equal training loss. (d) Validation loss.

Faster, and on a slightly different path. Muon reaches AdamW's final validation loss, 1.344, by step 7500 (d). Since they share the initialization and batches, if Muon just followed AdamW's path faster, they would be closer at equal loss (c) than at equal step (b). The opposite happens: from step 250 on, their distance is larger at equal loss than at equal step (at step 1000, 0.131 against 0.102). So the paths differ, but less than a change of data order does: from step 500 on, at equal step, AdamW and Muon on the same data order are closer than two data orders of one optimizer (a, b).

Endpoints. The final AdamW and Muon models are about as far apart as two runs of one optimizer on different data orders (b), but Muon ends at a lower validation loss: 1.309 against 1.344 (d).

Step 2: what differs along the way. We measure 30 properties of each model at 34 points of training, in five groups: weights W, one-step updates ΔW, loss curvature (Hessian H), internal activations and predictions (Appendix B defines each metric and compares all of them in Tables 3 and 4). Table 4 also gives the values for SGD with momentum. We compare the optimizers by each metric's ratio AdamW/Muon at training loss 1.81, on three data orders each (Fig. 2, Table 3). The ± gives its range over data orders; if it excludes 1, the optimizers differ more than data orders do.

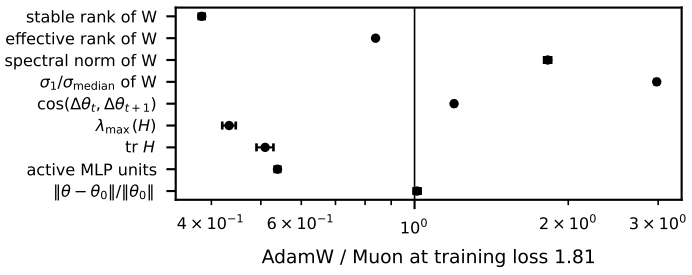


Figure 2: Ratio AdamW/Muon at training loss 1.81, for selected metrics; three data orders each; bars: range across data orders.

- *Weights W.* Muon's weight matrices have more evenly sized singular values. Their stable rank $\|W\|_F^2/\|W\|_2^2$ is 86 against AdamW's 33 (Muon/AdamW 2.61 ± 0.04), and their effective rank 325 against 272. Their spectral norm (2.9 against 5.4) and largest-to-median singular-value ratio $\sigma_1/\sigma_{\text{median}}$ (2.7 against 8.0) are lower. Both move similarly far from the initialization (ratio 1.01 ± 0.02): they differ in the shape of their weights, not in how far they travel.
- *One-step updates ΔW.* Consecutive Muon steps are less alike (cosine 0.74 against 0.88). Their stable rank is much higher (220 against 17; Table 3), but by construction: Muon orthogonalizes every update.
- *Loss curvature (Hessian H).* Muon is about twice as sharp: the top Hessian eigenvalue $\lambda_{\max}$ is 14 against 6.2, the trace $\operatorname{tr} H$ 2770 against 1400. This holds even though both take steps of similar relative size ($\|\Delta\theta\|/\|\theta\|$ is $6.53 \cdot 10^{-3}$ for Muon, $6.56 \cdot 10^{-3}$ for AdamW), so it is not an artifact of their different learning rates.
- *Internal activations.* AdamW's MLP is sparser: 12% of its units are active, against 22% for Muon.
- *Predictions.* No prediction metric separates the two: every AdamW/Muon range contains 1 (Table 3).

Step 3: swaps (Fig. 3). *Swaps.* A *swap* switches a run of one optimizer (the *parent*, on data order 0) to the other optimizer at a *fork step*, then trains it to step 10,000 on the parent's batches (Appendix C gives the fork steps, the optimizer state at the swap and the measurement steps). An optimizer's *own run* is its run from the start. We swap at nine fork steps from 50 to 4000, in both directions. We give each metric's value at step 10,000 as a fraction of the way from the parent's value (0) to the own run's (1).

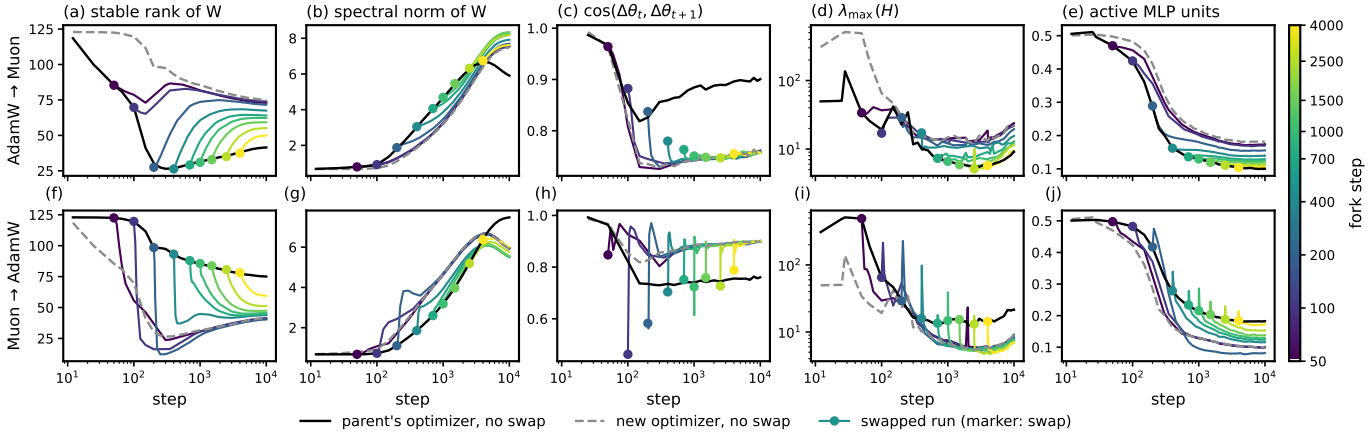


Figure 3: Swaps on data order 0: five metrics against the step. Top: AdamW to Muon; bottom: Muon to AdamW. Black: the parent's optimizer without a swap; grey dashed: the new optimizer's own run; colour: swapped runs, by fork step (dot: the swap).

Weights W and internal activations. For the spectral norm and stable rank of W (a, b, f, g) and the fraction of active MLP units (e, j), the final value depends on the fork step: the earlier the swap, the closer it ends to the new optimizer's own run. Even a swap at step 200 leaves a clear trace: from AdamW to Muon, the stable rank ends 0.90 of the way and the active fraction 0.67. This points to the importance of the early steps.

One-step updates ΔW. The cosine between consecutive updates moves to the new optimizer's level right after the swap (c, h). This is expected: it reflects the optimizer's own dynamics more than the model's weights.

Loss curvature (Hessian H). $\lambda_{\max}$ differs by direction (d, i). After a switch to AdamW, it falls to AdamW's level (1.00–1.18 of the way), often after a brief spike. After a switch to Muon, it behaves like the stable rank and active units above: the earlier the swap, the closer it ends to Muon's own run, and late swaps keep much of AdamW's value.

Next steps. (i) Repeat Steps 1–3 on a larger model (TinyStories-30M) and on character-level Shakespeare. (ii) For each property that differs in Step 2: does it follow directly from Muon's orthogonalized updates, or does it emerge during training? (iii) Do these differences explain Step 1, Muon's faster training and different path? Is any of them a reason Muon does better?

References

Eldan, R. and Y. Li (2023). *TinyStories: How Small Can Language Models Be and Still Speak Coherent English?*

Jordan, K., Y. Jin, V. Boza, et al. (2024). *Muon: An Optimizer for Hidden Layers in Neural Networks*. Blog post. URL: https://kellerjordan.github.io/posts/muon/.

Karpathy, A. (2022). *nanoGPT*. Software. URL: https://github.com/karpathy/nanoGPT.

A Hyperparameters

Table 1: Hyperparameters shared by the AdamW and Muon runs.

Runs	AdamW and Muon × data orders d0–d4 (10)
Model	nanoGPT, pre-LayerNorm blocks
Layers, heads, width	6, 6, 384
MLP	GELU, width 1536
Biases, dropout	none, none
Output head	untied
Attention projections	separate query, key, value
Context	256 tokens
Parameters	13.9M (10.6M without token and position embeddings and head)
Data	TinyStories (GPT-4 version)
Tokenizer	byte-level BPE, 4096 tokens, trained on the training split
Batch	128 windows × 256 tokens
Window offsets	from the training split; generator seeded with (data order, step), same for every optimizer
Steps	10,000
Tokens	328M (about 0.6 epochs)
Learning rate	$\eta\,\varphi(t)$ at step t, every parameter group
Warmup	linear, $\varphi(t) = (t+1)/200$ for $t < 200$
Decay	cosine to $\varphi = 0.1$ at step 10,000
Gradient clipping	none
Initialization seed	0, all runs
Training precision	bfloat16 autocast
Prediction, Hessian precision	float32, no autocast (TF32 matrix products allowed)
Logged predictions	next-token distributions: 128 fixed validation sequences × 16 evenly spaced positions (2,048), as float16 log-probabilities
Training loss	64 fixed training sequences, all positions; used to match runs at equal loss
Floor	mean JS between two data orders of one optimizer, at equal training loss; for two optimizers, the larger of their floors
Both logged at	every 250th step + 30 log-spaced steps in $[1, 10{,}000]$
Hessian	top eigenvalue, 20 Lanczos iterations, 32 fixed training sequences; AdamW also of $P^{-1/2}HP^{-1/2}$, with P AdamW's diagonal preconditioner
Measured at	every 500th step + 30 log-spaced steps

Table 2: Optimizer-specific hyperparameters. n_{out}, n_{in}: output and input size of the matrix.

	AdamW	Muon
η	$2 \cdot 10^{-3}$	0.01
η choice	best final validation loss of a 5-rate sweep on d0	as AdamW
Update	bias-corrected moments	Nesterov momentum; 5 quintic Newton–Schulz steps in bfloat16
Momentum	$\beta = (0.9, 0.95)$	$\beta = 0.95$
ϵ	10^{-8}	–
Update scale	–	$\sqrt{\max(1, n_{\text{out}}/n_{\text{in}})}$
Weight decay	0.1, decoupled; none on LayerNorm gains	none
Parameters	all	query, key, value, attention-output and both MLP matrices of every block
Aux. AdamW	–	embeddings, head, LayerNorm gains
Aux. η, β, ϵ	–	$2 \cdot 10^{-3}$, $(0.9, 0.95)$, 10^{-8}
Aux. weight decay	–	none

B Metrics

The Step-2 runs save checkpoints every 1000 steps, at 12 log-spaced steps, and at three consecutive steps $t, t+1, t+2$ for 24 values of t from 25 to 9998 (to measure one-step updates); we measure the metrics at 34 of these per run, and the Step-3 swaps at the steps in Table 5. *Weights*: for all parameters together, $\|\theta\|_2$, $\|\theta - \theta_0\|/\|\theta_0\|$ and $\|\theta\|_\infty/\|\theta\|_2$; and, averaged over the 36 weight matrices W of the blocks, the stable rank $\|W\|_F^2/\|W\|_2^2$, the effective rank $\exp H(\sigma/\sum\sigma)$ with H the entropy of the normalized singular values σ, the spectral norm, and $\sigma_1/\sigma_{\text{median}}$. *Updates* $\Delta\theta_t = \theta_{t+1} - \theta_t$: $\|\Delta\theta_t\|/\|\theta_t\|$, its cosine with $\Delta\theta_{t+1}$ and with the negative gradient $-g_t$ on step t's batch, and the stable and effective

rank of its block matrices ΔW. *Curvature* of the training loss on 32 fixed training sequences, with Hessian H: $\lambda_{\max}(H)$ (20 Lanczos iterations, logged during training), $\operatorname{tr} H$ (10 Rademacher probes), $\operatorname{tr} H/\lambda_{\max}$, and the batch sharpness $\mathbb{E}_B[g_B^\top H_B g_B]/\mathbb{E}_B\|g_B\|^2$, where g_B and H_B are the gradient and Hessian on a mini-batch B. *Predictions*, on the 2,048 held-out positions: entropy, mean top-1 probability, calibration error (15 bins), top-1 accuracy, loss on the targets in the top and the bottom third of training-set frequency and their difference, and JS to the unigram and to the bigram model of the training split. *Activations*, on 32 held-out sequences: attention entropy and attention on the first token (means over layers, heads and positions), fraction of MLP pre-activations above 0, RMS of the residual stream after the last block, and effective rank of the centred final features.

SGD, the reference in Table 4, uses heavy-ball momentum 0.9 and learning rate 0.06, the only rate tried between 0.03 and 0.12 that did not diverge on any of the three data orders; everything else is as in Table 1.

Table 3: All metrics of Step 2 at training loss 1.81: mean ± half-range over data orders 0–2, and the ratio AdamW/Muon. If AdamW's values range over $[a_{\min}, a_{\max}]$ and Muon's over $[b_{\min}, b_{\max}]$, the ratio ranges over $[a_{\min}/b_{\max},\ a_{\max}/b_{\min}]$, given as midpoint ± half-width. Rows marked ($\times 10^k$) are in units of 10^k. Bold: the five ratios farthest from 1 whose range excludes 1.

	AdamW	Muon	A/M
Weights W			
weight norm $\|\theta\|_2$	224 ± 2	224 ± 0	0.997 ± 0.013
distance from init. $/\|\theta_0\|$	2.18 ± 0.03	2.15 ± 0.01	1.01 ± 0.02
largest entry $/\|\theta\|_2$ ($\times 10^{-3}$)	7.04 ± 0.06	6.45 ± 0.02	1.09 ± 0.01
stable rank of W	33.0 ± 0.3	86.2 ± 0.3	$\mathbf{0.383 \pm 0.005}$
effective rank of W	272 ± 0	325 ± 0	0.839 ± 0.000
spectral norm of W	5.39 ± 0.07	2.95 ± 0.01	1.82 ± 0.03
$\sigma_1/\sigma_{\text{median}}$ of W	8.01 ± 0.05	2.68 ± 0.00	$\mathbf{2.98 \pm 0.03}$
One-step updates ΔW			
step size $\|\Delta\theta\|/\|\theta\|$ ($\times 10^{-3}$)	6.56 ± 0.08	6.53 ± 0.03	1.01 ± 0.02
stable rank of ΔW	16.7 ± 0.3	220 ± 0	$\mathbf{(7.66 \pm 0.14) \times 10^{-2}}$
effective rank of ΔW	228 ± 1	369 ± 0	0.618 ± 0.003
cosine of consecutive steps	0.883 ± 0.002	0.738 ± 0.003	1.19 ± 0.01
cosine of step and $-g$	0.244 ± 0.004	0.217 ± 0.007	1.12 ± 0.06
Loss curvature (Hessian H)			
top Hessian eigenvalue	6.18 ± 0.10	14.3 ± 0.2	$\mathbf{0.433 \pm 0.013}$
batch sharpness	0.974 ± 0.254	3.13 ± 0.37	$\mathbf{0.332 \pm 0.116}$
Hessian trace	1400 ± 40	2770 ± 20	0.510 ± 0.019
trace / top eigenvalue	224 ± 6	196 ± 7	1.16 ± 0.08
Internal activations			
attention entropy (nats)	2.51 ± 0.03	2.79 ± 0.02	0.899 ± 0.014
attention on first token ($\times 10^{-2}$)	1.75 ± 0.11	1.98 ± 0.05	0.908 ± 0.079
active MLP units	0.120 ± 0.001	0.222 ± 0.001	0.539 ± 0.007
residual-stream RMS	4.73 ± 0.05	3.38 ± 0.02	1.40 ± 0.03
effective rank of features	312 ± 2	320 ± 0	0.977 ± 0.007
Predictions			
prediction entropy (nats)	1.89 ± 0.01	1.88 ± 0.01	1.01 ± 0.01
mean top-1 probability	0.561 ± 0.003	0.563 ± 0.001	0.994 ± 0.007
calibration error ($\times 10^{-2}$)	3.25 ± 0.17	3.05 ± 0.46	1.09 ± 0.22
top-1 accuracy	0.546 ± 0.005	0.544 ± 0.005	1.01 ± 0.02
loss on frequent targets	0.896 ± 0.004	0.892 ± 0.008	1.00 ± 0.01
loss on rare targets	3.10 ± 0.02	3.10 ± 0.01	1.00 ± 0.01
rare minus frequent loss	2.21 ± 0.02	2.21 ± 0.01	0.996 ± 0.015
JS to unigram model	0.800 ± 0.001	0.802 ± 0.001	0.998 ± 0.002
JS to bigram model	0.469 ± 0.001	0.471 ± 0.001	0.996 ± 0.004

Table 4: The metrics of Table 3 with SGD as a reference: ratios AdamW/SGD and Muon/SGD. Bold: in each ratio column, the five ratios farthest from 1 whose range excludes 1.

	AdamW	Muon	SGD	A/S	M/S
Weights W					
weight norm $\|\theta\|_2$	224 ± 2	224 ± 0	110 ± 0	2.04 ± 0.02	2.05 ± 0.01
distance from init. $/\|\theta_0\|$	2.18 ± 0.03	2.15 ± 0.01	0.486 ± 0.006	4.47 ± 0.11	4.42 ± 0.07
largest entry $/\|\theta\|_2$ ($\times 10^{-3}$)	7.04 ± 0.06	6.45 ± 0.02	39.5 ± 1.0	0.180 ± 0.006	$\mathbf{0.165 \pm 0.005}$
stable rank of W	33.0 ± 0.3	86.2 ± 0.3	56.2 ± 5.6	0.614 ± 0.069	1.60 ± 0.17
effective rank of W	272 ± 0	325 ± 0	321 ± 1	0.849 ± 0.003	1.01 ± 0.00
spectral norm of W	5.39 ± 0.07	2.95 ± 0.01	1.47 ± 0.11	3.61 ± 0.31	1.98 ± 0.15
$\sigma_1/\sigma_{\text{median}}$ of W	8.01 ± 0.05	2.68 ± 0.00	4.37 ± 0.55	1.79 ± 0.23	0.599 ± 0.074
One-step updates ΔW					
step size $\|\Delta\theta\|/\|\theta\|$ ($\times 10^{-3}$)	6.56 ± 0.08	6.53 ± 0.03	0.624 ± 0.047	$\mathbf{10.9 \pm 1.0}$	$\mathbf{10.8 \pm 0.9}$
stable rank of ΔW	16.7 ± 0.3	220 ± 0	8.37 ± 0.48	2.04 ± 0.15	$\mathbf{26.6 \pm 1.6}$
effective rank of ΔW	228 ± 1	369 ± 0	205 ± 2	1.12 ± 0.02	1.81 ± 0.02
cosine of consecutive steps	0.883 ± 0.002	0.738 ± 0.003	0.857 ± 0.004	1.03 ± 0.01	0.863 ± 0.007
cosine of step and $-g$	0.244 ± 0.004	0.217 ± 0.007	0.447 ± 0.010	0.550 ± 0.022	0.491 ± 0.027
Loss curvature (Hessian H)					
top Hessian eigenvalue	6.18 ± 0.10	14.3 ± 0.2	50.2 ± 0.6	$\mathbf{0.123 \pm 0.004}$	0.284 ± 0.008
batch sharpness	0.974 ± 0.254	3.13 ± 0.37	6.38 ± 0.50	$\mathbf{0.162 \pm 0.051}$	0.504 ± 0.095
Hessian trace	1400 ± 40	2770 ± 20	$(2.55 \pm 0.04) \times 10^4$	$\mathbf{(5.55 \pm 0.25) \times 10^{-2}}$	$\mathbf{0.109 \pm 0.002}$
trace / top eigenvalue	224 ± 6	196 ± 7	507 ± 9	0.447 ± 0.020	0.385 ± 0.021
Internal activations					
attention entropy (nats)	2.51 ± 0.03	2.79 ± 0.02	3.84 ± 0.01	0.655 ± 0.009	0.728 ± 0.007
attention on first token ($\times 10^{-2}$)	1.75 ± 0.11	1.98 ± 0.05	1.82 ± 0.07	0.972 ± 0.095	1.07 ± 0.07
active MLP units	0.120 ± 0.001	0.222 ± 0.001	0.262 ± 0.002	0.458 ± 0.008	0.850 ± 0.012
residual-stream RMS	4.73 ± 0.05	3.38 ± 0.02	0.244 ± 0.003	$\mathbf{19.4 \pm 0.5}$	$\mathbf{13.8 \pm 0.3}$
effective rank of features	312 ± 2	320 ± 0	297 ± 2	1.05 ± 0.01	1.07 ± 0.01
Predictions					
prediction entropy (nats)	1.89 ± 0.01	1.88 ± 0.01	1.94 ± 0.00	0.971 ± 0.006	0.966 ± 0.007
mean top-1 probability	0.561 ± 0.003	0.563 ± 0.001	0.558 ± 0.001	1.00 ± 0.01	1.01 ± 0.00
calibration error ($\times 10^{-2}$)	3.25 ± 0.17	3.05 ± 0.46	2.97 ± 0.28	1.08 ± 0.16	1.02 ± 0.25
top-1 accuracy	0.546 ± 0.005	0.544 ± 0.005	0.554 ± 0.003	0.987 ± 0.013	0.980 ± 0.014
loss on frequent targets	0.896 ± 0.004	0.892 ± 0.008	0.851 ± 0.012	1.05 ± 0.02	1.05 ± 0.02
loss on rare targets	3.10 ± 0.02	3.10 ± 0.01	3.11 ± 0.04	0.999 ± 0.018	0.999 ± 0.015
rare minus frequent loss	2.21 ± 0.02	2.21 ± 0.01	2.26 ± 0.03	0.979 ± 0.020	0.982 ± 0.017
JS to unigram model	0.800 ± 0.001	0.802 ± 0.001	0.804 ± 0.001	0.995 ± 0.002	0.997 ± 0.003
JS to bigram model	0.469 ± 0.001	0.471 ± 0.001	0.462 ± 0.001	1.02 ± 0.00	1.02 ± 0.00

C Swaps

Table 5: Swaps. A: parent's optimizer; B: optimizer after the swap; s: fork step.

Parents	the d0 runs of AdamW and Muon (Table 1)
Warm state	the other optimizer, stepped on the same mini-batch gradients with learning rate 0: momenta, second moments and step counts evolve as if training; parameters follow A alone
Fork steps s	50, 100, 200, 400, 700, 1000, 1500, 2500, 4000
Saved at s	parameters; states of both optimizers
Swap $A \to B$ loads	parameters at s; B's warm state
Hyperparameters	B's (Table 2)
Steps	s to 10,000, on the parent's schedule and batches (data order 0)
Swap forks	2 directions $\times$ 9 steps = 18
Metrics (Fig. 3)	at $s + 0$, 5, 10, 25, 50, 100, 250, 500, 1000, 2000, 4000, 6000 (each with the next two steps, for the updates) and every 1000 steps
Same-batch control	the parent itself (A continued with the same batches)